

PARTH MAHENDRA PURI

AI ENGINEER

B.E. Artificial Intelligence & Data Science (AI & DS)

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AI Engineer with a strong foundation in **Machine Learning, Deep Learning, Generative AI, and Agentic AI**, experienced in building **production-grade LLM pipelines, and Responsible AI governance** delivering impact across **MRO, healthcare, and insurance domains**.

SKILLS

- **Programming Languages:** Python.
- **AI & Machine Learning:** Artificial Intelligence, Machine Learning, Deep Learning, Generative AI, Agentic AI, LLM Fine-tuning (SFT, PEFT/LoRA), RAG (Semantic & Text-to-SQL), Prompt Engineering, Computer Vision, NLP.
- **Models & APIs:** OpenAI GPT-4.1 / GPT-5 / GPT-5.4, Anthropic Claude 3.7 Sonnet / Sonnet 4.5 / 4.6 / Opus 4.6, Google Gemini 2.5 Flash / 2.5 Pro / 3.0 Flash / 3.1 Pro, LLaMA (3.2-3B & 1B), Mistral (3B & 7B), DeepSeek (7B, 8B, 14B), BERT, YOLO, LSTMs, GRUs, ResNet-50, AlexNet, VGG19.
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, MCP (Model Context Protocol), FastAPI, LangChain, LangGraph, Hugging Face, Unsloth, Streamlit.
- **Cloud & DevOps:** AWS (EC2, S3, Lambda, Bedrock, SageMaker AI, Guardrails), GCP (Vertex AI), Docker, Git & GitHub, Linux.
- **Databases & Infrastructure:** PostgreSQL, MySQL, MongoDB, Redis, Vector Store (FAISS, ChromaDB), System Architecture.

EXPERIENCE

- Artificial Intelligence (AI) Engineer at Verdantis (Ultria, Verdantis), Mumbai.** **Mar 2025 - Present**
- Developed enterprise AI systems leveraging frontier generative models **OpenAI GPT-4.1 / GPT-5 / GPT-5.4, Anthropic Claude 3.7 Sonnet / Sonnet 4.5 / 4.6 / Opus 4.6, and Google Gemini 2.5 Flash / 2.5 Pro / 3.0 Flash / 3.1 Pro** to deliver scalable, production-grade solutions.
 - Designed **Agentic Workflows** and end-to-end **Generative AI pipelines** for autonomous, **context-aware task execution** and developed robust **AI Prediction using Generative AI**, increasing SMEs (Subject Matter Experts) work efficiency in the MRO industry by **95%**.
 - Implemented **AI Governance** by designing Workflows with **strict Human-in-the-Loop (HITL) validation gates**, ensuring Experts verified all automated outputs before deployment, Engineered **Responsible AI** by **reducing hallucinations by 40%** and **preventing the generation of non-factual corporate data**.
 - Built and deployed 4 production Agentic AI Modules in MRO 360 integrated with SAP via PostgreSQL & MongoDB : **Obsolete Parts, Work Order, Spare Part Criticality, Reorder Point** using **LangChain, LangGraph & Gemini**, integrated with SAP via **PostgreSQL & MongoDB** achieving **97% accuracy and improving operational profitability by 40%**.
- Machine Learning Engineer at Health India Insurance TPA Pvt Ltd, Mumbai.** **Nov 2024 - Mar 2025**
- Developed and trained **Machine Learning and Deep Learning models** to automate processes and enhance operational efficiency **70%**.
 - Built **data extraction pipelines** leveraging **Generative AI** and worked extensively with **Large Language Models (LLMs)**, focusing on **fine-tuning LLMs and hyperparameter optimization**, achieving over **91% accuracy in UAT testing**.
 - Worked with various **pre-trained models** like **BERT, Llama (3.2-3B & 1B), Mistral (3B & 7B), Deepseek (7B, 8B, 14B), YOLO, LSTMs, GRUs**, as well as applications in **Image Processing, Azure OCR, document reconstruction, text classification, and transformers**. Also Built **RAG pipelines with MongoDB and Langchain**. Ensured **PII security and compliance** by deploying an **on-premise pipeline with zero external data transfer**.
- AI Engineer Intern at Sharv Auto Tech Solutions Private Limited, Mumbai.** **May 2024 - Nov 2024**
- Enhancing **Natural Language Processing capabilities** using **Large Language Models (LLMs)** and **Retrieval-Augmented Generation (RAG)**, including **finetuning LLMs**. Working with **SQL Database and Fast API** for backend development.
- AI Engineer Intern at Fafadia Tech, Mumbai.** **May 2023 - Apr 2024**
- Worked on **Pretrained Models, Similarity Modeling, and Transfer Learning**. Developed **Machine Learning and Deep Learning Models** including **Auto-encoders and Convolutional Neural Networks (CNNs)**, for **advanced feature extraction** from large datasets, **reducing search latency by 40%**, significantly enhancing user experience and system performance.

PROJECTS

- Research Assistant Agent** **Jan 2026 - Mar 2026**
- Architected a fully **autonomous, end-to-end research pipeline** using **LangGraph** with a **Supervisor-Worker multi-agent architecture**, implementing a **self-evaluating feedback loop** that dynamically assesses **research quality in real-time** and **autonomously triggers refined search iterations with zero manual intervention**. Leveraged **Agentic Function Calling** for intelligent task decomposition and dynamic workflow routing, enabling **goal-directed autonomous execution** across specialized worker agents.
 - Engineered **100% a production-grade FastAPI async REST API with PostgreSQL** for stateful agent orchestration at scale.

EDUCATION

- B.E. in Artificial Intelligence and Data Science.** **Jun 2020 - May 2024**
- University of Mumbai, Vidyavardhini's College Engineering and Technology, Mumbai - 8.04 CGPA

CERTIFICATION AND PUBLICATION

- Published Research Paper named "Retroflex: Uncovering Visual Equivalences through Reverse Image Recon," 2024 11th INDIACom, New Delhi, India, 2024. [IEEE.explore](#) **Mar 2024**
- Software Engineer Intern, HackerRank. [HackerRank](#) **Feb 2024**
- Data Science Intern, Mumbai Metropolitan Region Development Authority (MMRDA). **Jun 2022 - Jul 2022**
- Generative AI on GCP: Deep Dive Into Generative AI Studio. [InfosysSpringBoard](#) **Aug, 2024**
- Machine Learning with TensorFlow (InfosysSpringBoard). **Jul, 2024**